

Reproduction of huggingface/lighteval: SmolLM2-1.7B-Instruct on ARC-Challenge and HellaSwag

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Independent reproduction · Benchmark or leaderboard entry · Natural language processing

Abstract

Ran the README evaluation with lighteval at the pinned commit (accelerate backend):
SmolLM2-1.7B-Instruct on the full ARC-Challenge test set with 25-shot prompts and the full HellaSwag validation set with 10-shot prompts, batch size 8. At this commit HellaSwag is scored generatively (exact match) rather than by log-likelihood accuracy. A fresh cache directory was used so every sample was computed in the sealed run. One NVIDIA RTX 5060 Ti.

About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], which remain authoritative.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	SmolLM2-1.7B ARC-C and H...	2026-10-08	1	1	Signed by the K-Veritas server

2 Results

r1: SmolLM2-1.7B ARC-C and HellaSwag

Run	Metric	Final value	Step
1	hellaswag_em	0.409679	
1	arc_challenge_acc	0.508532	

Run	Inline claim	Value	Phase
1	hellaswag_em	0.409679	
1	arc_challenge_acc	0.508532	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	4204.9 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	f67c1daa6e72d2157c461f19858ffdf82a40d62056e09d47555bdf70ed2e29d	7.0 MB
r1.kvbundle.zip	16bb5e71f3a93b8a24e69d162d2bb61b90615373edd473bae11f476c9de2f8d6	21.5 MB

5 Links and licence

Reproduction of: <https://github.com/huggingface/lighteval/tree/7fa19dc6f53a713ba8bb6541e7ed66dad2e24fd7>

Record: CC-BY 4.0. Code: MIT, as declared by the authors. Tags: llm-evaluation, arc, hellaswag, smollm2.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (SmolLM2-1.7B ARC-C and HellaSwag), kv:2610.00025v1/r1. Data hash 89ce7eee4602d9d203d3f3986b0360b2646900cf6a05df02b9b18608839cc520.

<https://kveritas.org/records/2610.00025v1/r1.pdf>